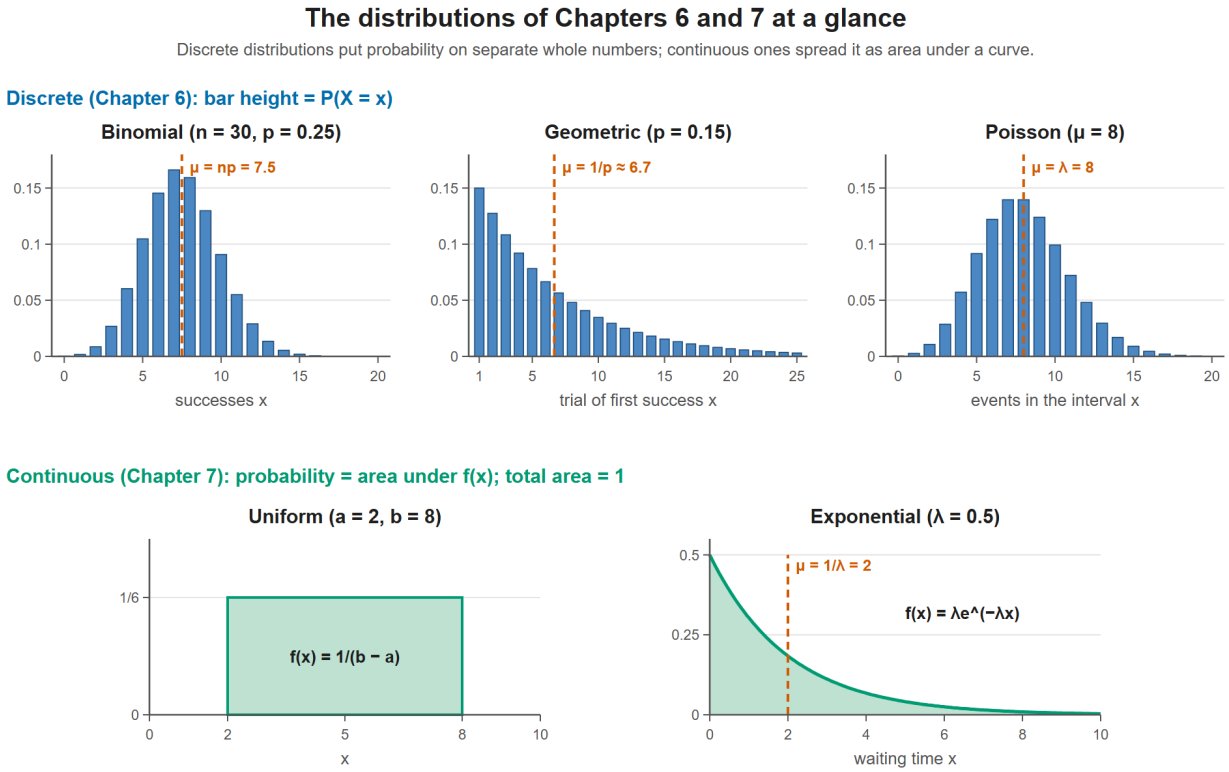
Lesson 6.1 - Probability Distribution Formulas

> **Video:** Probability Distribution Formulas by The Organic Chemistry Tutor **Chapter:** 6 - Discrete Probability Distributions

A reference sheet for the distributions in Chapters 6 and 7. Throughout, p = probability of success and $q = 1 - p$ = probability of failure.



What each distribution looks like. The three discrete ones (top) give probability as the height of a bar at a whole number. The two continuous ones (bottom) give probability as the area under the curve, and the total area is always 1. The dashed lines mark each mean.

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1. Binomial distribution

X = number of successes in n independent trials.

$$P(X = x) = {}_nC_x\, p^x q^{n-x}, \qquad {}_nC_x = \frac{n!}{(n-x)!\,x!}$$

Mean	Standard deviation
$\mu = np$	$\sigma = \sqrt{npq}$

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2. Geometric distribution

X = the trial on which the **first success** happens.

$$P(X = n) = q^{n-1}p$$

Why: for the first success to be on trial 4, trials 1-3 must fail ($q \cdot q \cdot q$) and trial 4 must succeed (p), so $P(X = 4) = q^3p$.

Probability	Formula
$P(X > n)$	q^n
$P(X \geq n)$	q^{n-1}
$P(X \leq n)$	$1 - q^n$
$P(X < n)$	$1 - q^{n-1}$

Mean	Variance	Standard deviation
$\mu = \frac{1}{p}$	$\sigma^2 = \frac{1}{p}\left(\frac{1}{p} - 1\right) = \mu(\mu - 1) = \frac{q}{p^2}$	$\sigma = \frac{\sqrt{1-p}}{p}$

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3. Poisson distribution

X = number of events in a fixed interval, where the average number of events is $\mu = \lambda$.

$$P(X = n) = \frac{\mu^n e^{-\mu}}{n!} = \frac{\lambda^n e^{-\lambda}}{n!}$$

Probability	Formula
$P(X \leq n)$	$e^{-\mu}\sum_{x=0}^{n}\frac{\mu^x}{x!}$
$P(X > n)$	$1 - e^{-\mu}\sum_{x=0}^{n}\frac{\mu^x}{x!}$

Mean	Variance	Standard deviation
$\mu = \lambda$ ($= np$ when approximating a binomial)	$\sigma^2 = \lambda$	$\sigma = \sqrt{\lambda}$

> In the video the Poisson denominator is spoken as "n squared". The correct denominator is $n!$ (n factorial), as in the other form written on screen.

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4. Uniform distribution (continuous)

X is equally likely to be anywhere between a and b. The graph is a **rectangle** of width $b - a$, and its total area must be 1:

$$\text{area} = \text{base} \times \text{height} = (b - a) \cdot f(x) = 1 \;\Rightarrow\; f(x) = \frac{1}{b - a}$$

Probability = partial width ÷ total width:

$$P(c \leq X \leq d) = \frac{d - c}{b - a}$$

For example, $P(a \le X \le c) = \dfrac{c-a}{b-a}$ and $P(c \le X \le b) = \dfrac{b-c}{b-a}$.

Mean	Standard deviation
$\mu = \dfrac{a+b}{2}$	$\sigma = \dfrac{b-a}{\sqrt{12}}$

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5. Exponential distribution (continuous)

Waiting times. λ = rate parameter, and $\mu = 1/\lambda$.

$$f(x) = \lambda e^{-\lambda x}, \quad x \ge 0$$

Probability	Formula	From calculus
$P(X \le n)$ (left area)	$1 - e^{-\lambda n}$	$\displaystyle\int_0^n \lambda e^{-\lambda x}\,dx$
$P(X \ge n)$ (right area)	$e^{-\lambda n}$	$\displaystyle\int_n^\infty \lambda e^{-\lambda x}\,dx$

Mean	Variance	Rate
$\mu = \dfrac{1}{\lambda}$	$\sigma^2 = \dfrac{1}{\lambda^2}$	$\lambda = \dfrac{1}{\mu}$

In general, a probability for a continuous distribution is the **area under** $f(x)$: $P(a \le X \le b) = \int_a^b f(x)\,dx$.

Where each one is used

Distribution	Typical question
Binomial	"Exactly / at least x successes in n trials" (Lesson 6.2)
Geometric	"First success on trial n" (Lesson 6.3)
Poisson	"Number of events in a time or space interval" (Lesson 6.4)
Uniform	"Equally likely anywhere between a and b" (Lesson 7.3)
Exponential	"Waiting time until the next event" (Lesson 7.4)

Lesson 6.2 - Finding the Probability of a Binomial Distribution

> **Video:** Finding The Probability of a Binomial Distribution Plus Mean & Standard Deviation by The Organic Chemistry Tutor **Chapter:** 6 - Discrete Probability Distributions

Formula

$$P(X = x) = {}_nC_x \, p^x \, q^{n-x}, \qquad {}_nC_x = \frac{n!}{(n-x)!\,x!}$$

Symbol	Meaning
n	number of trials
x	number of successes you want (**exactly** x)
p	probability of success on one trial
$q = 1 - p$	probability of failure on one trial

Mean and standard deviation: $\mu = np, \quad \sigma = \sqrt{npq}$

Conditions: a fixed number of trials, each trial is success or failure, the trials are independent, and p is the same on every trial.

Example 1 - rolling a die

A die is rolled **12** times. What is the probability of getting a **4 exactly five times**?

- $n = 12$, $x = 5$, $p = \frac{1}{6}$ (one face out of six), $q = \frac{5}{6}$

Combination by hand:

$$_{12}C_5 = \frac{n!}{(n-x)!\,x!} = \frac{12!}{7!\,5!} = \frac{12 \cdot 11 \cdot 10 \cdot 9 \cdot 8}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 11 \cdot 9 \cdot 8 = 792$$

Probability:

$$P(X = 5) = {}_nC_x \, p^x \, q^{n-x} = 792 \left(\tfrac{1}{6}\right)^5 \left(\tfrac{5}{6}\right)^7 \approx 0.0284 = 2.84\%$$

Example 2 - guessing on a multiple-choice test

A test has **20** questions, each with 4 choices (A-D) and one correct answer. A student guesses on every question. What is the probability of **exactly 6** correct?

- $n = 20$, $x = 6$, $p = \frac{1}{4} = 0.25$, $q = 0.75$

$$_{20}C_6 = \frac{n!}{(n-x)!\,x!} = \frac{20!}{14!\,6!} = \frac{20 \cdot 19 \cdot 18 \cdot 17 \cdot 16 \cdot 15}{6!} = 38{,}760$$

$$P(X = 6) = {}_nC_x \, p^x \, q^{n-x} = 38{,}760 \, (0.25)^6 (0.75)^{14} \approx 0.1686 = 16.86\%$$

(With 5 answer choices instead, $p = 0.20$ and $q = 0.80$.)

Example 3 - students taking algebra

25% of students at a school take algebra. **30** students are chosen at random.

- $n = 30$, $p = 0.25$, $q = 0.75$

(a) P(exactly 7 take algebra)

$$P(X = 7) = {}_nC_x \, p^x \, q^{n-x} = {}_{30}C_7 \, (0.25)^7 (0.75)^{23} = 2{,}035{,}800 \, (0.25)^7 (0.75)^{23} \approx 0.1662 = 16.6\%$$

(b) P(fewer than 5 take algebra), so $X < 5$, meaning $X = 0, 1, 2, 3, 4$ (5 is not included):

x	$P(X = x) = {}_{30}C_x (0.25)^x (0.75)^{30-x}$
0	0.000179
1	0.001786
2	0.008631
3	0.026853
4	0.060420
Sum	**0.09787 ≈ 9.79%**

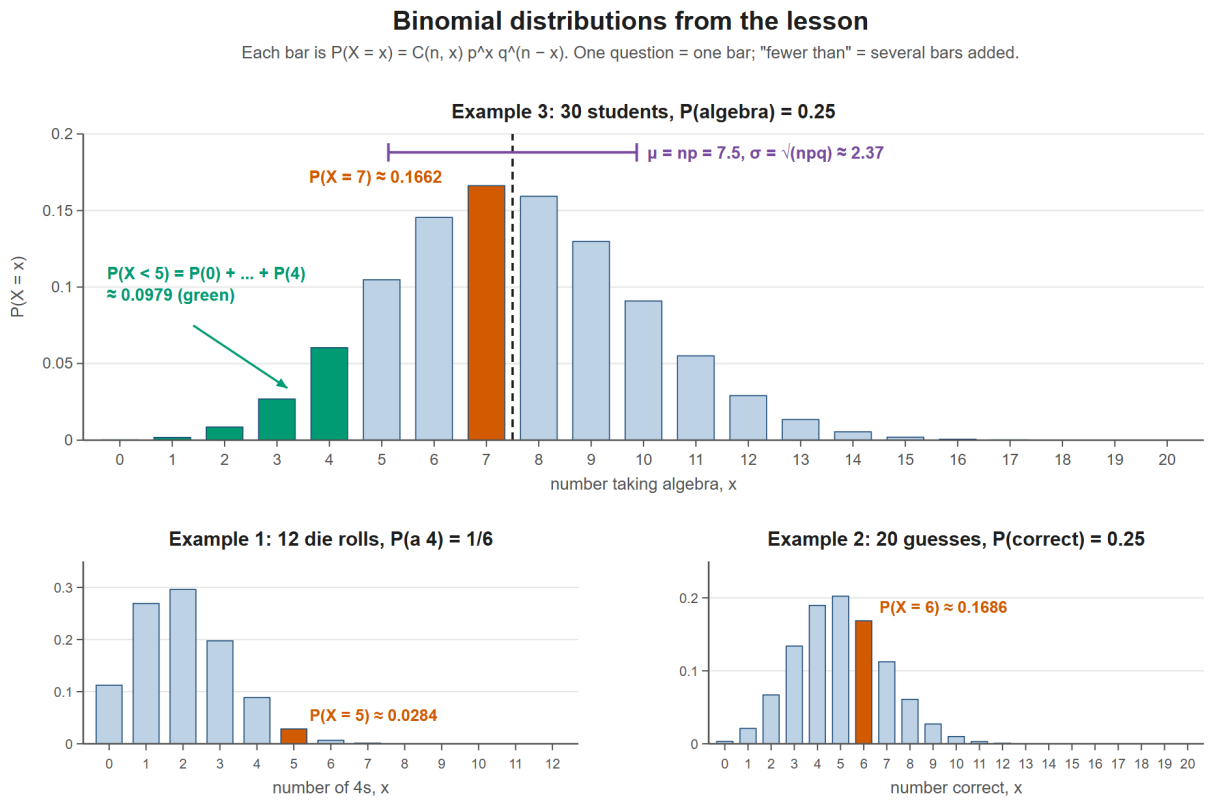
> **Correction:** the video adds only $x = 1$ to 4 and gets 9.77%. "Fewer than 5" also includes $x = 0$ (no algebra students at all), which adds 0.00018, so the correct total is about **9.79%**. The difference is tiny here, but leaving out $x = 0$ is a common mistake.

For large n, adding many terms gets tedious. A normal approximation, or the complement rule with a calculator's binomial CDF, is faster.

(c) Mean and standard deviation

$$\mu = np = 30(0.25) = 7.5$$

$$\sigma = \sqrt{npq} = \sqrt{30(0.25)(0.75)} = \sqrt{5.625} \approx 2.37$$



The full distribution behind each example. "Exactly 7" is a single bar. "Fewer than 5" is the sum of the five green bars, including the almost invisible bar at x = 0. The bars are tallest near the mean np = 7.5.

Key takeaways

- Identify n, x, p and q first, then substitute into ${}_nC_x p^x q^{n-x}$.
- "Exactly x" → one term. "Fewer than / at most / at least" → **add several terms**, and remember $x = 0$ when it applies.
- $\mu = np$ and $\sigma = \sqrt{npq}$.

Lesson 6.3 - The Geometric Distribution

> **Video:** Geometric Distribution - Probability, Mean, Variance, & Standard Deviation by The Organic Chemistry Tutor **Chapter:** 6 - Discrete Probability Distributions

The idea

A **geometric distribution** answers: *on which trial does the **first** success happen?*

If the first success is on trial x, then the first $x-1$ trials failed and trial x succeeded:

$$P(X=x)=q^{x-1}p, \qquad q=1-p$$

All the formulas

Question	Formula
First success **on** trial x	$P(X=x)=q^{x-1}p$
First success **by** trial x (within the first x)	$P(X\le x)=1-q^{x}$
First success **before** trial x	$P(X<x)=1-q^{x-1}$
First success **after** trial x	$P(X>x)=q^{x}$
First success on trial x **or later**	$P(X\ge x)=q^{x-1}$
Mean (expected trial of the first success)	$\mu=\dfrac{1}{p}$
Variance	$\sigma^2=\dfrac{1}{p}\left(\dfrac{1}{p}-1\right)=\dfrac{1-p}{p^2}$
Standard deviation	$\sigma=\sqrt{\dfrac{1-p}{p^2}}=\dfrac{\sqrt{1-p}}{p}$

Why $P(X>x)=q^x$: "no success in the first x trials" means x failures in a row. Then $P(X\le x)=1-q^x$ is its complement.

Tip for $<$ and $\ge$: shift by one so you can use the formulas above. $P(X<7)=P(X\le 6)$ and $P(X\ge 18)=P(X>17)$.

Example 1 - first six on the fourth roll

$p=\frac{1}{6}$ (rolling a 6), $q=\frac{5}{6}$:

$$P(X=4)=q^{x-1}p=\left(\tfrac{5}{6}\right)^3\cdot\tfrac{1}{6}=\frac{125}{1296}\approx 0.0965=9.6\%$$

Three failures, then one success.

Example 2 - first blue car is the 7th car

15% of cars are blue: $p=0.15$, $q=0.85$.

$$P(X=7)=q^{x-1}p=(0.85)^6(0.15)\approx 0.0566=5.7\%$$

Example 3 - the 10th person you meet is the first teacher

4% of a town's residents are teachers: $p = 0.04$, $q = 0.96$.

(a)

$$P(X = 10) = q^{x-1}\,p = (0.96)^{9}(0.04) \approx 0.0277 = 2.77\%$$

(b) Mean, variance, standard deviation

$$\mu = \frac{1}{p} = \frac{1}{0.04} = 25, \qquad \sigma^2 = \frac{1}{p}\left(\frac{1}{p} - 1\right) = 25(25-1) = 600, \qquad \sigma = \sqrt{\sigma^2} = \sqrt{600} \approx 24.49$$

Example 4 - defective tires ($p = 0.02$, $q = 0.98$)

2% of tires are defective, and a sample of tires is tested one at a time.

(a) The 8th tire is the first defective one

$$P(X = 8) = q^{x-1}\,p = (0.98)^{7}(0.02) \approx 0.0174 = 1.7\%$$

(b) The first defect is found within the first 5 tires Adding each case:

$$P(X \le 5) = p + qp + q^2p + q^3p + q^4p = p(1 + q + q^2 + q^3 + q^4) \approx 0.0961$$

With the cumulative formula, which is much faster:

$$P(X \le 5) = 1 - q^{x} = 1 - (0.98)^{5} \approx 0.0961 = 9.6\%$$

(c) The first defect is found within the first 10

$$P(X \le 10) = 1 - q^{x} = 1 - (0.98)^{10} \approx 0.1829 = 18.3\%$$

(d) How many tires do you expect to test to find the first defect?

$$\mu = \frac{1}{p} = \frac{1}{0.02} = 50 \text{ tires}$$

Example 5 - more cumulative questions (same tires)

(a) First defect before the 7th tire

$$P(X < 7) = P(X \le 6) = 1 - q^{6} = 1 - (0.98)^{6} \approx 0.1142 = 11.4\%$$

(b) First defect after the 12th tire

$$P(X > 12) = q^{x} = (0.98)^{12} \approx 0.7847 = 78.5\%$$

(c) First defect on the 18th tire or later

$$P(X \ge 18) = P(X > 17) = (0.98)^{17} \approx 0.7093 = 70.9\%$$

(d) First defect between tire 15 and tire 30 (inclusive)

Split the whole range (total probability 1) into three pieces: A = $X \le 14$, B = $15 \le X \le 30$, C = $X > 30$.

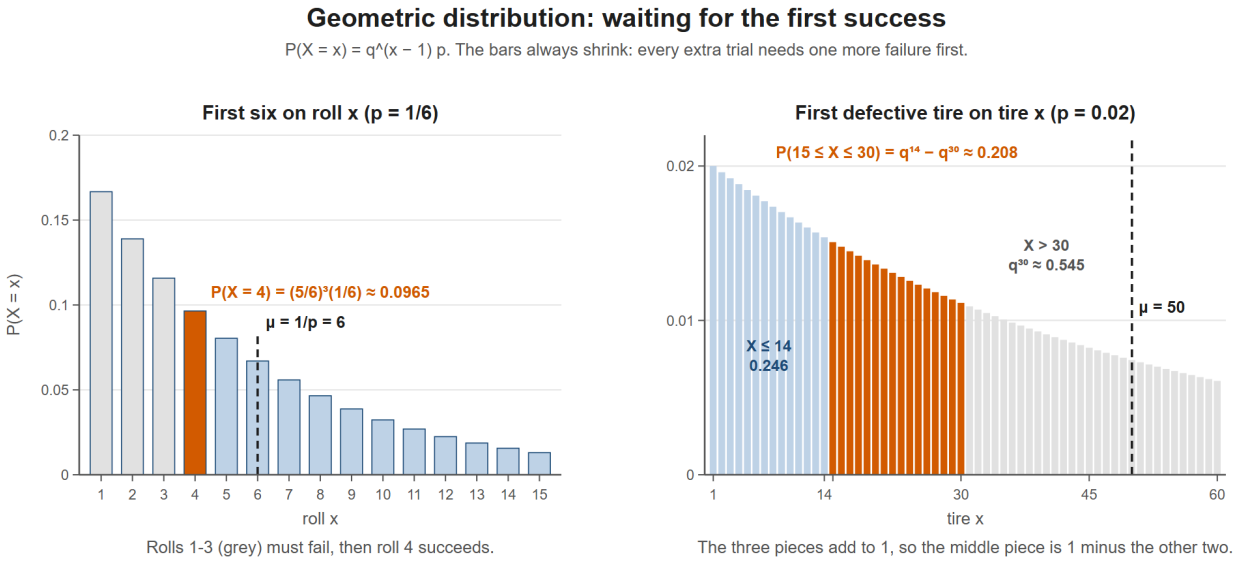
$$P(15 \le X \le 30) = 1 - P(X \le 14) - P(X > 30)$$

$$= 1 - \left[1 - (0.98)^{14}\right] - (0.98)^{30} = (0.98)^{14} - (0.98)^{30} \approx 0.7536 - 0.5455 = 0.2082 = 20.8\%$$

(e) First defect after tire 20 but no later than tire 45 ($21 \le X \le 45$)

$$= 1 - P(X \leq 20) - P(X > 45) = (0.98)^{20} - (0.98)^{45} \approx 0.6676 - 0.4029 = 0.2647 = 26.5\%$$

> **Shortcut:** $P(a \leq X \leq b) = q^{a-1} - q^{b}$. Check: (d) is $q^{14} - q^{30}$ ✓ and (e) is $q^{20} - q^{45}$ ✓. Make sure no trial is counted twice or left out.



Left: the first six on roll 4 needs three failures (grey) and then a success. Right: for the tires, the range 15 to 30 (orange) is whatever is left after removing X ≤ 14 and X > 30, which is q^14 − q^30.

Key takeaways

- Geometric = waiting for the **first** success: $P(X = x) = q^{x-1}p$.
- Cumulative: $P(X \leq x) = 1 - q^{x}$ and $P(X > x) = q^{x}$. Convert $<$ and $\geq$ by shifting by 1.
- $\mu = 1/p$ is the expected number of trials until the first success.

Lesson 6.4 - The Poisson Distribution

> **Video:** Introduction to Poisson Distribution - Probability & Statistics by The Organic Chemistry Tutor
> **Chapter:** 6 - Discrete Probability Distributions

The idea

The **Poisson distribution** gives the probability of **exactly x events in a fixed interval** (of time, area, ...) when the events happen at a known **average rate** μ (also written λ).

$$P(X = x) = \frac{\mu^x \, e^{-\mu}}{x!}$$

- $e \approx 2.71828$
- $x! = x(x-1)\cdots 2 \cdot 1$, with $0! = 1$

The mean must match the interval

If the interval changes, scale the mean in proportion:

$$\frac{\mu_2}{\mu_1} = \frac{t_2}{t_1}$$

Example 1 - customers per day

A business averages **12 customers per day**. What is the probability of exactly **8** customers in one day?

$$P(X = 8) = \frac{\mu^x \, e^{-\mu}}{x!} = \frac{12^8 e^{-12}}{8!} \approx 0.0655 = 6.55\%$$

Example 2 - text messages

A student receives on average **7 texts per 2 hours**.

(a) Exactly 9 texts in 2 hours. The interval is the same, so $\mu = 7$:

$$P(X = 9) = \frac{\mu^x \, e^{-\mu}}{x!} = \frac{7^9 e^{-7}}{9!} \approx 0.1014 = 10.14\%$$

(b) Exactly 24 texts in 8 hours. The interval is 4 times longer, so the mean is 4 times larger:

$$\frac{\mu_2}{7} = \frac{8}{2} \;\Rightarrow\; \mu_2 = 28$$

$$P(X = 24) = \frac{\mu^x \, e^{-\mu}}{x!} = \frac{28^{24} e^{-28}}{24!} \approx 0.0601 = 6.01\%$$

Example 3 - phone calls ($\mu = 8$ per hour)

(a) Exactly 7 calls in one hour

$$P(X = 7) = \frac{\mu^x \, e^{-\mu}}{x!} = \frac{8^7 e^{-8}}{7!} \approx 0.1396 = 13.96\%$$

(b) At most 5 calls ($X \leq 5$, so $X = 0, 1, \ldots, 5$)

Factor out $e^{-\mu}$, since only $\mu^x / x!$ changes from term to term:

$$P(X \leq 5) = e^{-8}\left(\frac{8^0}{0!} + \frac{8^1}{1!} + \frac{8^2}{2!} + \frac{8^3}{3!} + \frac{8^4}{4!} + \frac{8^5}{5!}\right)$$

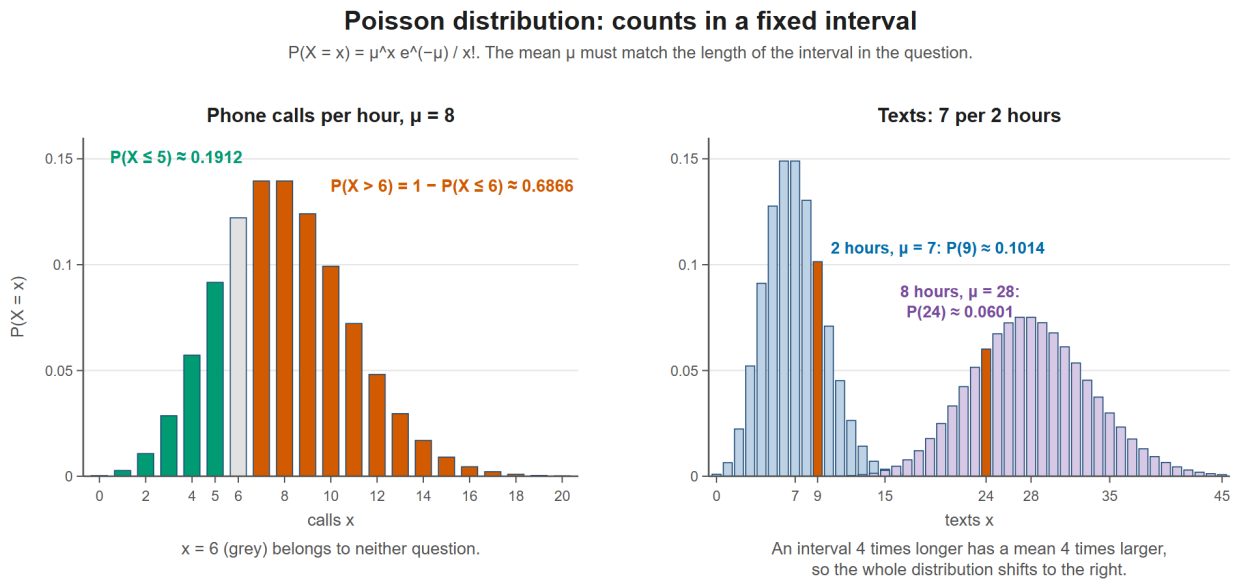
$$= e^{-8}\left(1 + 8 + 32 + 85.33 + 170.67 + 273.07\right) = e^{-8}(570.07) \approx 0.1912 = 19.12\%$$

(c) More than 6 calls ($X > 6$, which runs on to infinity)

Use the complement, since the total probability is 1:

$$P(X > 6) = 1 - P(X \leq 6) = 1 - e^{-8}\sum_{x=0}^{6}\frac{8^x}{x!} = 1 - e^{-8}(934.16) \approx 1 - 0.3134 = 0.6866 = 68.7\%$$

This is likely, which makes sense: the average is 8 calls, so more than 6 is the usual case.



Left: "at most 5" adds the green bars and "more than 6" is everything orange, found as 1 − P(X ≤ 6). Right: changing the interval from 2 hours to 8 hours multiplies the mean by 4, which moves and spreads out the whole distribution.

Key takeaways

- Poisson: $P(X = x) = \dfrac{\mu^x e^{-\mu}}{x!}$ counts events in a fixed interval.
- **Adjust** μ whenever the interval in the question differs from the one in the given rate.
- "At most k" → add $x = 0$ to k. "More than k" → $1 - P(X \leq k)$.
- Mean = variance = μ, and the standard deviation is $\sqrt{\mu}$.

Lesson 6.5 - Continuous vs Discrete Data

> **Video:** Continuous vs Discrete Data by The Organic Chemistry Tutor **Chapter:** 6 - Discrete Probability Distributions

Definitions

	Continuous data	Discrete data
Values	**Any** value in an interval	Only **specific** values (often whole numbers)
How it's obtained	**Measured**	**Counted**
Values between 1 and 10	1, 1.3, 2.46, 5.389, ...	1, 2, 3, ..., 10
Example	Temperature: 83.6 °F, 99.46 °F	Rolling a die: 1-6 only (never 2.37)

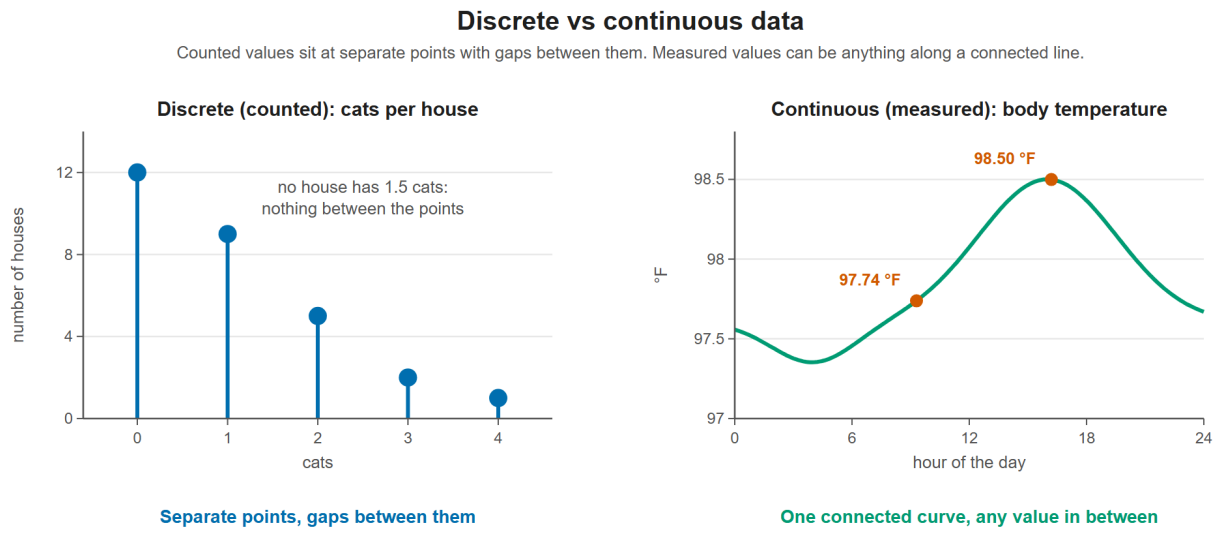
Practice - continuous (C) or discrete (D)?

#	Quantity	Answer	Reason
1	Speed of a car	**C**	Measured: 25.6 mph, 42.86 mph, ...
2	Number of cats in a neighborhood	**D**	Counted; no half cats
3	Your body weight	**C**	Measured: 160 lb, 173.4 lb, ...
4	Number of doctors in a town	**D**	Counted
5	Number of books on a shelf	**D**	Counted
6	Height of a student	**C**	Measured: 5.84 ft, 6.3 ft, ...
7	Number of students in a classroom	**D**	Counted
8	Your age **rounded** to the nearest whole number	**D**	Only whole numbers: 16, 19, 22, 24
9	Your **exact** age	**C**	Measured: 16.2, 19.3, 21.76, 23.958 years

Items 8 and 9 show that **rounding** a continuous quantity turns it into discrete data.

Recognizing them on a graph

- **Continuous:** one connected line or curve with no jumps or gaps.
- **Discrete: separate points**, or bars/steps that don't connect into a smooth line.



Counted data (left) can only sit at whole numbers, so the graph is separate points. Measured data (right) can take any value, so the graph is one connected curve.

Key takeaways

- **Measured** → **continuous** (any value in a range). **Counted** → **discrete** (specific values only).
- Rounding turns continuous data into discrete data.
- Graphs: a smooth connected line means continuous; isolated points mean discrete.
- This distinction decides which distributions apply: binomial, geometric and Poisson are **discrete** (Chapter 6); uniform, exponential and normal are **continuous** (Chapters 7-8).